

Technical Task Brief

Intern Work Plan · 6 Weeks

Indian Institute of Technology Kharagpur

Duration 6-week engagement starting [date to be confirmed]

Reporting One-page written update submitted every Friday by 6 PM

Contact PI (contact details shared separately)

Repository Access granted on Day 1

Team at a Glance

#	Intern	Primary Role	Core Deliverables (6 Weeks)
1	Sahil Kumar Gupta	Backend + Doctor/Admin Portal + RAG	DB migration · Alert engine · Gemini RAG fallback · Doctor dashboard APIs · Admin portal · Cloud Run deployment
2	Saugata Malakar	Healthcare Privacy + LLM + Multimodal AI	Data anonymisation (DPDP Act 2023) · Federated learning PoC · Multimodal AI integration · Clinical NLP
3	Adreesh Mitra	Wound Image Processing & Segmentation	OpenCV preprocessing · SAM2 wound segmentation · Coin-based area measurement · Wound boundary AI
4	Sharif Hossain Sarkar	Wound AI Model Training & Deployment	Wagner Grade EfficientNet-B0 · Wound colour CNN · TFLite export · Flask inference endpoints
5	Kousttav Paul	Skin AI + Mobile App Features	EfficientNet-B3 skin model · Leprosy recall $\geq 95\%$ · React Native screen fixes · ASHA training modules
6	Shivraj Gulve	Eye AI + RAG + Risk Scoring + Deployment	MobileNetV3 eye triage · Diabetic risk scoring · LlamaIndex RAG · pgvector · Streamlit analytics

Sharif Hossain Sarkar | Wound AI Model Training & Deployment

Your stack	PyTorch / TensorFlow, EfficientNet-B0 (timm), Google Colab Pro A100, Weights & Biases, TFLite, ONNX, Flask
You are responsible for	Two wound AI models: wound severity classification and wound tissue type analysis. The inference endpoint that packages them, and the rigorous model evaluation needed for publication.
First-day priority	Share W&B project access with the analytics engineer on Day 1, his model dashboard reads your training logs from the start. Also confirm with the CV engineer that you will use his dataset split, not a new one.
Semester	Continuation strongly recommended. Fine-tuning on locally collected patient data is the highest-value post-launch activity. Colab-async, feasible alongside coursework.

What you are building

Two wound AI models. The severity classifier tells the monitoring system how serious a wound is. The tissue type model detects whether active infection is present. Both feed into the same inference endpoint, which routes through the backend system with Gemini Vision as a fallback when confidence is low. The evaluation you run in Week 5, with proper held-out test sets and 95% confidence intervals, is the data that a wound monitoring accuracy paper is built on. Run it rigorously.

Week by week

Week	Theme	What to do	What to hand in
Week 1	Data & environment	- Use the dataset split the CV engineer saves to the shared Drive, do not create a different split, or your validation numbers will not align with his CV pipeline paper - Build the DataPipeline class: 224×224 resize, ImageNet normalisation, augmentation (rotation $\pm 30^\circ$, brightness $\pm 20\%$, zoom 0.8–1.2×, horizontal flip, Gaussian noise). Log the class distribution for all wound severity grades. Imbalanced classes need weighted loss - Set up Colab Pro with A100, Drive checkpoints, and Weights & Biases. Share the W&B project with the analytics engineer on Day 1, his model dashboard reads your training logs	- DataPipeline class (unit tested) - Class distribution chart - W&B project live and shared
Week 2	Wound severity classifier	- Architecture: EfficientNet-B0, ImageNet pretrained (timm library). Freeze backbone, train head 5 epochs, unfreeze top 20% of backbone, fine-tune 15–20 more epochs - Loss: weighted cross-entropy. Weight it so that under-grading is penalised more than over-grading, missing a severe grade has worse clinical consequences than flagging a mild one as moderate - Targets: $\geq 75\%$ top-1 accuracy overall; the most clinically critical grade classes should individually hit $\geq 85\%$. Export TFLite (float16) and ONNX	- Wound severity model ($\geq 75\%$ top-1 accuracy) - Confusion matrix + per-class AUROC - TFLite + ONNX exports verified - W&B run logged
Week 3	Wound tissue CNN	- Four tissue classes: granulation (healthy), slough (stalled healing), eschar (necrosis), cellulitis (active infection). Images come from the CV engineer's segmented sub-regions, not the full photograph - Asymmetric loss for cellulitis and eschar. A missed active infection is clinically dangerous. Cellulitis sensitivity must hit 90% on the held-out test set, not just validation - Also build a secondary binary classifier for periwound spreading redness (redness extending beyond the wound margin, this is a cellulitis indicator even when the wound itself looks contained)	- Wound tissue CNN ($\geq 85\%$ accuracy, $\geq 90\%$ cellulitis sensitivity on test set) - Periwound binary classifier - POST /infer/wound endpoint live (wound severity + tissue type + periwound output)

Week 4	Inference endpoint & deployment	- Package the wound pipeline: CV preprocessing → SAM2 segmentation → severity model → tissue model → structured JSON out. Register with the main Flask inference router - JSON output must include: severity_grade, grade_confidence, tissue_colour, colour_confidence, periwound_redness, wound_area_cm2, fallback_triggered - Batch inference: one monitoring session involves three photographs. The endpoint must handle all three in one call. Latency target: ≤6 seconds on CPU. Test the Gemini fallback trigger on a low-confidence image	- POST /infer/wound live (all JSON fields, batch inference) - Latency benchmarked (≤6s) - Gemini fallback confirmed on low-confidence input
Week 5	Evaluation for publication	- Run full evaluation on the held-out test set, the 15% you never touched during training. Per-class sensitivity, specificity, PPV, NPV, and AUROC, all with 95% confidence intervals (Wilson method). This is what goes in the wound monitoring paper - Calibration plot: does a confidence of 0.8 actually mean right 80% of the time? Document any miscalibration - Write model cards for both models. Include architecture, training data, test-set metrics, known failure modes (model will underperform on very dark eschar and on dark skin tones, document this honestly), and the recommended confidence threshold - Export evaluation data as a CSV for the analytics engineer, agree the column names with him before Week 5	- Evaluation report (per-class sensitivity/specificity/AUROC, 95% CI) - Model card, wound severity model - Model card, wound tissue model - Eval CSV for analytics engineer (agreed column format)
Week 6	Harden & hand over	- Quantise TFLite models to float16. Measure size reduction and check accuracy drop is small. Stress test: 100 random images, zero crashes, zero memory leaks, all confidence scores in [0, 1] - Write the fine-tuning protocol for when local patient image data becomes available. What format does it expect? How long does a fine-tuning run take? What accuracy improvement would 500 additional images provide? Document this so the next person does not have to figure it out	- TFLite float16 benchmarks (size, latency, accuracy delta) - Stress test report (100 images, 0 crashes) - Fine-tuning protocol document

Deliverables

#	Deliverable	By	Submit where
1	DataPipeline class + class distribution charts	Wk 1	GitHub + Drive
2	Wound severity model (≥75% top-1) + TFLite + ONNX	Wk 2	GitHub + Drive
3	Wound tissue CNN (≥85% acc, ≥90% cellulitis sensitivity on test)	Wk 3	GitHub + Drive
4	POST /infer/wound (batch inference, Gemini fallback tested)	Wk 4	GitHub + Swagger
5	Evaluation report (95% CI, per-class AUROC)	Wk 5	Drive doc
6	Wound severity + tissue model cards	Wk 5	Drive docs
7	Eval CSV for analytics engineer (agreed format)	Wk 5	Drive folder
8	TFLite float16 benchmarks + stress test report	Wk 6	Drive doc
9	Fine-tuning protocol document	Wk 6	Drive doc

Weekly Report Template

One page. Submitted every Friday by 6 PM. The PI will respond by Monday. *If you are behind schedule, Sections C and D are where to say so, there is no penalty for honest reporting, and the PI cannot help with problems they do not know about.*

Heath Screening AI Project Weekly Progress Report	
One page · Friday by 6 PM · Specific numbers and links, not summaries	
Intern name:	
Role:	
Week number and date:	
GitHub username:	
W&B or Drive link this week:	
A. Work completed this week (4–5 bullets)	
Example format (replace these with your own answers): - Completed SAM2 wound segmentation module. Tested on 50 images from the dataset, IoU 0.73, above the 0.70 target. - Fixed CameraScreen crash on Android API 28 (null permission result handler). Closed GitHub Issue #31. - Wound severity model fine-tuning complete: 78.4% top-1 on validation set. Target was 75%, so we are through. - Deployed POST /fieldworker/ask endpoint. Tested 20 sample questions. 18 of 20 returned relevant answers. Your answer:	
B. Code and files submitted (exact links, not 'see GitHub')	
Example format (replace these with your own answers): - GitHub PR #47, SAM2 segmentation module, branch feat/wound-seg: github.com/[repo]/pull/47 - Drive, dataset split and class distribution charts: drive.google.com/[link] - W&B run, wound severity model v3 (run ID ws_v3): wandb.ai/[link] Your answer:	
C. Problems faced (specific, what broke and what you tried)	
Example format (replace these with your own answers): - SAM2 IoU drops to 0.41 on high-eschar wounds (low contrast). Trying stronger CLAHE pre-processing this week. - Leprosy class has only 143 images after DermNet NZ download. Oversampling to 800 with augmentation, not yet sure if that is enough for 95% sensitivity. - Gemini API rate limit hit during batch testing. Will throttle to smaller batches. Your answer:	

D. Help needed from PI (who, what, by when)

Example format (replace these with your own answers):

- *Need the dataset download credentials from PI by Monday, wound severity training is blocked.*
- *Need NHA sandbox client_secret by Wednesday, health ID integration (Flow A) is blocked.*
- *Requesting backend engineer code review on database migration script by Tuesday.*

Your answer:

E. Targets for next week (3–4 measurable goals, include numbers)

Example format (replace these with your own answers):

- *Wound severity model to $\geq 75\%$ top-1 on validation. Export TFLite and ONNX by Friday.*
- *Wire SAM2 output into tissue CNN, POST /infer/wound returns severity grade and tissue type in one response.*
- *Close all 7 Functional-severity mobile bugs in GitHub Issues by Thursday.*

Your answer:

Self-assessment: On track Slightly behind Need urgent help (circle one)

PI notes: *(returned by Monday)*